

A Practical Indoor Positioning System based on Collaborative PDR and Wi-Fi Fingerprinting

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Abstract—An Indoor Positioning System (IPS) is vital in providing location-based service (LBS) indoors. The shortcomings of the Global Positioning System (GPS) indoors have led to the development of many alternative solutions for IPS. Wi-Fi fingerprinting is a promising solution due to its ability to accurately predict indoor locations. Moreover, the widespread deployment of Wi-Fi infrastructures makes the Wi-Fi fingerprinting-based IPS technique readily applicable. However, this technique requires high overhead due to the need to define many reference points (RPs) within the subject area. Multiple Wi-Fi received signal strength (RSS) samples shall be collected from each RP, introducing extensive manual work. This fact makes the Wi-Fi fingerprinting-based IPS less practical. This issue is addressed by employing pedestrian dead reckoning (PDR) as part of the proposed technique. Realizing a practical IPS based on collaborative use of PDR and Wi-Fi fingerprinting. In the proposed method, most of the time, the location prediction will be based on PDR. When the step count reaches a threshold, a switch from PDR to Wi-Fi fingerprinting is made to decide the user's location. Then, the system resets the step count and uses the PDR again. This approach significantly reduces the total number of RPs needed by the system. The numerical results show that the proposed technique has 99% lower overhead than the conventional Wi-Fi fingerprinting technique and achieves 55.6% better prediction accuracy than the standalone PDR technique. The results justify the benefits of the proposed IPS based on collaborative PDR and Wi-Fi fingerprinting.

Index Terms—indoor positioning system, PDR, Wi-Fi fingerprinting, low overhead, high accuracy

I. INTRODUCTION

Global Positioning System (GPS) is a technology commonly used to track the position of a person or an asset. GPS works by transmitting signals from satellites orbiting the Earth to receiving devices on the ground. However, the effectiveness of GPS heavily relies on the ability of the satellite signals to propagate to the receiving devices. During propagation, satellite signals encounter obstacles such as buildings, mountains, vegetation, etc., weakening the signal strength and thereby reducing GPS accuracy [1]. This makes GPS unsuitable for use as a location-determining tool in indoor areas.

The Indoor Positioning System (IPS) is a system that can provide location information, which can be utilized for

asset tracking, building navigation, and enhancing customer experiences in retail. Considering that the Indoor Positioning System (IPS) is vital in supporting indoor location-based services, an IPS technique with high accuracy is required. There are various types of IPS, each with its characteristics, using different technologies such as Wi-Fi, Bluetooth Low Energy (BLE), Radio-Frequency Identification (RFID), Ultra-Wideband (UWB), and Infrared (IR) [2]. Wi-Fi has a particular advantage among these technologies due to its widespread deployment in the existing indoor infrastructure. This widespread deployment can be leveraged to support the implementation of high-accurate IPS [3].

Wi-Fi-based IPS generally uses the fingerprinting technique, which involves creating a database of Wi-Fi signal characteristics throughout the grid of indoor locations that comprise the IPS coverage area. Devices match real-time RSSI data from an object with fingerprint data in the database to determine the position [4]. Although the accuracy level of the fingerprinting technique is very high, the overhead of the fingerprinting process becomes an obstacle to its real-world implementation. To develop a more practical Wi-Fi-based IPS using fingerprinting, the overhead of fingerprint data collection must be reduced without significantly lowering the accuracy level.

This paper proposes a design of an IPS based on collaborative Pedestrian Dead Reckoning (PDR) and Wi-Fi fingerprinting. The PDR can complement Wi-Fi-based IPS by tracking a person's movements through step counting and direction estimation using inertial sensors on a smartphone device. By integrating fingerprinting with PDR, the aim is to develop a system that reduces the need to collect Wi-Fi fingerprints excessively while maintaining high prediction accuracy. The location prediction will be based on PDR most of the time. Due to the nature of PDR, the cumulative error of PDR increases with the number of steps. Therefore, the system uses a threshold to determine when to switch from PDR to Wi-Fi fingerprinting to minimize the prediction error. We study the behavior of PDR to propose the appropriate threshold values. We have implemented the proposed IPS and evaluated its performance under realistic scenarios. All of those become the contribution of this work.

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II. RELATED WORKS

Researchers have done extensive studies regarding IPS. The works in [5], [6], and [7] provide comprehensive surveys about the latest technology and algorithm for IPS. Authors in [6] present a review of various IPS that utilize landmarks for indoor navigation and location tracking. In the landmark-based IPS, specific areas are identified and used as a reference point for determining the location. Due to its unique characteristics, the landmark-based IPS can achieve higher accuracy than the standard method. Moreover, the paper also discusses the integration of sensors, data fusion techniques, and algorithmic advancements that enhance positioning precision in complex indoor environments. However, the landmark-based IPS faces a challenge when the environment changes dynamically. On the other hand, paper [7] summarizes the technologies used for IPS, including satellite-based, magnetic-based, inertial sensors, sound-based, optical-based, vision-based, and radio-frequency. The inertial sensors and radio frequency-based IPS are reported to have high accuracy but also high complexity at the same time. The PDR and WiFi RSSI-based IPS are examples of inertial sensors and radio frequency-based IPS, respectively. Because of their high accuracy, these two methods are a promising solution for high-precision IPS. Then, researchers compete to develop a technique to cope with their drawbacks and make them practical.

In [8], the Authors explore the possibility of using WiFi fingerprinting and PDR technique to accurately determine the node's location. The location prediction is performed in two steps. The first step uses the WiFi RSSI to get the distance of each node to the access point. The second step utilizes the PDR technique (accelerometer sensor data) to get the user movement. Then, the results of those two techniques are combined using a sensor fusion approach to decide the node's location. However, the calculation of distance based on RSSI may be inaccurate, specifically in highly dynamic radio frequency environments, making the system less accurate. Furthermore, the Authors in [9] propose a method that combines PDR and Wi-Fi fingerprinting. These two methods result in the k-node (WiFi access point) participating in the KNN fingerprint algorithm. Based on the selected k-node, the IPS decides the user's location by running the K-NN fingerprinting matching Algorithm. This work does not address the overhead issue introduced using a WiFi signal fingerprinting map. The introduced overhead may cause the developed IPS to be impractical due to the high cost required for building the fingerprinting map.

On the other hand, the work in [10] presents an improved integrated IPS based on WiFi fingerprinting and PDR. The main goal is to enhance the accuracy of IPS. The Authors state that the main reason for the inaccurate prediction of the WiFi fingerprinting method is the fluctuation of radio signals in indoor environments. The proposed system employs a three-step processing technique to reduce this adverse effect. In the first step, the proposed method considers the indoor structure, such as corners, to improve the prediction. Then, the system incorporates a traditional weighted K-nearest neighbor to weaken the effect of WiFi signal fluctuations. Finally, the particle filter (PF) technique integrates the WiFi fingerprinting, PDR, and indoor map to produce a high-accuracy result. Meanwhile, the Authors in [11] proposed

a collaborative WiFi fingerprinting and PDR-based IPS. Initially, the system independently estimates the user's location based on RSSI and PDR. Then, the system combines the results from those two methods by weighting the contribution of each technique to the final result. The system utilizes the adaptive extended Kalman Filter (AEKF) to dynamically configure the weight value considering the changes in the indoor environment and sensor signal variations. The above two techniques require the full WiFi fingerprinting map of the subject location to be made in advance. The Authors must collect 20000 WiFi fingerprint information for a relatively small area to enable their system to operate. This number is huge considering that in practical conditions, much larger areas shall be considered, such as the entire building of the University, Government Complex, Shopping mall, etc.

III. COLLABORATIVE WiFi FINGERPRINTING AND PDR

The problem with traditional WiFi fingerprinting-based IPS is that the WiFi fingerprint shall be collected at each location on the map. The grid approach is usually used for this purpose. In this approach, the grid is drawn over the map. Then, the administrator collects the WiFi fingerprint for each grid. The grid where WiFi fingerprints are collected is referred to as a reference point (RP). The process of collecting WiFi fingerprints can be costly when a large area is considered. Thus, the heuristic behind the design is to reduce the effort of creating a WiFi fingerprint map as much as possible. To realize this, the use of PDR must be maximized to avoid the need to depend on the availability of WiFi fingerprints on every grid on the map. As a result, WiFi fingerprints do not need to be collected on each grid. Using only this approach introduces a problem *i.e.*, the longer PDR is used, the higher the error that will be obtained. This is because in each step, the PDR introduces an error, and the error is accumulated since the current location prediction will be used to predict the next position. The following subsection discusses the efforts to mitigate the problem.

A. PDR Usage Threshold

As mentioned earlier, the use of PDR introduces cumulative error. Thus, to keep the prediction accuracy high, The point at which to stop using the PDR needs to be decided and switch to WiFi fingerprinting for prediction. A steps threshold is introduced (*StepThres*), which is defined as the maximum number of steps that can be depended only on PDR. While the steps are less than this threshold from the starting point, The prediction error is expected to be under an acceptable value.

Upon stopping using PDR, WiFi fingerprinting will be used to predict the location. At this point, an RP location close to the current position can be expected. Then, the WiFi Fingerprinting technique can map the location to that RP. The predicted location by WiFi fingerprints will become the next starting point for PDR prediction. any WiFi fingerprinting-based prediction algorithm can be applied to decide the grid location of the user.

B. Reference Point Location

The availability of RP is essential in keeping the prediction accuracy high. The more RPs are in the system, the more accurate the prediction will be. However, as more RPs will increase overhead and make the system impractical, the

number of RPs should be kept as small as possible while maintaining high prediction accuracy. For this purpose, the RP should be an area where there is a high possibility that users will step or pass closely over it. Moreover, the distance between RPs shall be decided with respect to the total travel distance of users while using PDR only. For instance, if the $StepThres = 25$ and assuming every step is 40 cm, the total travel distance while using PDR only is 10 m. This implies that every 10m, a switch is made from PDR to WiFi fingerprinting for location prediction.

Thanks to the indoor structure of the building, deciding such a location of RP is trivial. First, the location that is very certain will be passed by the user as RPs can be tagged, such as room door, lift door, or stair door. Then, for each room, one or more RPs can be set depending on the size of the room. To start, a center point of the room can be selected as RP and then each corner of the room is chosen as additional RP, if needed. Finally, RPs are selected within the corridor based on the rule that the distance between RPs shall be less than the total travel distance using PDR only, as explained above.

C. Fusion Algorithm

The Algorithm 1 presents the combined WiFi fingerprinting and PDR technique.

Algorithm 1 Collaborative PDR and Wi-Fi Fingerprinting

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1: Set  $StepThres$ ,  $ScanThres$ , and  $DistanceThres$ 
2: Perform initial scan
3: while true do
4:    $stepCount \leftarrow 0$ ,  $scanAttempt \leftarrow 0$ 
5:   while  $stepCount \leq StepThres$  do
6:      $\mathcal{P}_p \leftarrow$  Use PDR to determine location
7:      $stepCount \leftarrow stepCount + 1$ 
8:   end while
9:   while  $scanAttempt \leq ScanThres$  do
10:    Perform WiFi Scanning
11:     $\mathcal{P}_f \leftarrow$  Location (WiFi fingerprinting)
12:     $distance \leftarrow$  Euclidean( $\mathcal{P}_p, \mathcal{P}_f$ )
13:    if  $distance \leq DistanceThres$  then
14:      break
15:    end if
16:     $scanAttempt \leftarrow scanAttempt + 1$ 
17:  end while
18:  if  $scanAttempt > ScanThres$  then
19:    Perform initial scan
20:  end if
21: end while

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1) *Initial Scan*: In the initial scan (Lines 2 and 20), the system determines the location using WiFi fingerprinting in the following steps. First, the application will request three different location predictions, with each prediction request using RSS data from different scans. Note that each prediction request returns the top 3 data points. Thus, the application obtains nine different data points, each represented as follows:

$$Data = (x_i, y_i, z_i, c_i) \quad (1)$$

Here, $p_i = (x_i, y_i, z_i)$ represent the coordinates data point and c_i represents the confidence level of a prediction i . In determining an accurate location, several variables must be considered, namely the confidence values and the occurrence

of predictions across multiple trials. Therefore, the confidence counts data for each unique coordinate and the occurrence count of those unique coordinates in the nine data points are required. The confidence count data is expressed as follows:

$$CS(p_j) = \sum_{i=1}^n c_i \cdot \delta(p_i, p_j) \quad (2)$$

While the occurrence count data is expressed as follows:

$$C(p_j) = \sum_{i=1}^n \delta(p_i, p_j) \quad (3)$$

These two data points are multiplied to produce a combined score determining the best prediction candidate, which is normalized to obtain the probability value. To obtain the normalization value, the total confidence count and the total occurrence count are required, expressed as follows:

$$TCS = \sum_j CS(p_j) \quad (4)$$

$$TC = \sum_j C(p_j) \quad (5)$$

Thus, each unique coordinate has a normalized score expressed as follows:

$$NP(p_j) = \frac{CS(p_j) \times C(p_j)}{TCS \times TC} \quad (6)$$

After obtaining the normalized scores for each unique coordinate, the user's initial location will be determined using the coordinate with the highest normalized score.

2) *Prediction using PDR*: After determining the initial position, the system starts using PDR to track the user's movements. PDR utilizes the accelerometer and magnetometer sensors on the smartphone to estimate the user's position based on their steps and direction of movement. PDR will produce the total number of steps and the final position of the user's movement.

The system monitors the user's step count when PDR is running. If the number of steps taken is still below the threshold, the system continues to use PDR to update the user's position without performing a new Wi-Fi scan (Line 5–8). This is done to maximize the accuracy of the position prediction. If the step count exceeds the threshold, a combination algorithm of the scan with the final PDR position is executed. The step threshold is based on the number of steps that result in significant errors during PDR testing.

3) *Wi-Fi Scan and PDR Combination Algorithm*: Overall, this system uses PDR as the primary positioning tool. Then, it uses Wi-Fi fingerprinting to determine the initial location and calibrate the PDR position results.

The system performs a Wi-Fi scan when the step count exceeds the specified $StepThres$. The results of the Wi-Fi scan are used to obtain location predictions from the machine learning service. These predictions are then combined with the final position calculated by PDR to update the user's position. The calculation is done by finding the Euclidean distance between the prediction results and the final PDR position. The closest position from the prediction results will be adjusted to determine whether it falls under the category of near or far based on the distance threshold. This is done to ensure that errors do not worsen. This combination helps reduce propagation errors that may occur in PDR over time.

This process repeats continuously until the distance between the final PDR position and the prediction results falls below the *DistanceThres* or reaches the *ScanThres*. *DistanceThres* is the distance used as the limit to decide when position correction is needed. When the Euclidean distance between the PDR coordinates and the Wi-Fi fingerprinting prediction results is less than the *DistanceThres*, the user's position is updated according to the Wi-Fi fingerprinting results. On the other hand, *ScanThres* is the number of scans set as the limit before re-performing the initial scan. As the number of steps increases, the resulting error worsens, so the correction process using Wi-Fi fingerprinting will not yield a good result. Thus, the user's new position will be re-determined using the initial scan algorithm. This is done because Wi-Fi fingerprinting is more reliable than PDR in this condition.

IV. IMPLEMENTATION

A. System Architecture

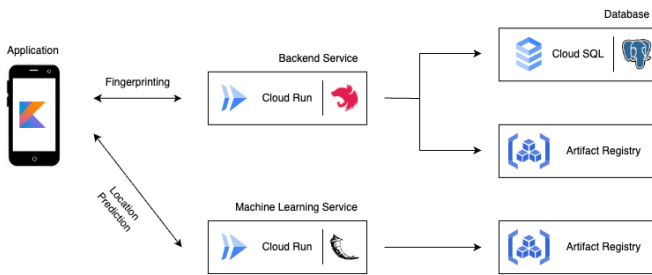


Fig. 1. System architecture

As illustrated in the system architecture Fig. 1, the system adopts the microservices concept. This approach aims to reduce inter-service dependencies, allowing each service to be independently scaled and deployed. The system is divided into three major parts: client-side, backend service, and machine learning service. The client-side is an Android application. The backend service is crucial in managing and storing the required data. The machine learning service stores various machine learning models trained using offline data.

B. Wi-Fi Fingerprinting Implementation

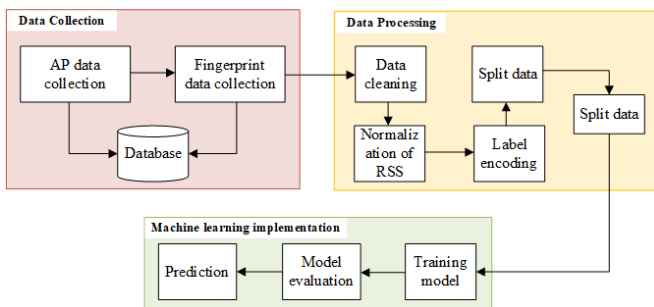


Fig. 2. Wi-Fi fingerprinting implementation

Fig. 2 explains how the Wi-Fi fingerprinting implementation flow works. The flow consists of data (Wi-Fi fingerprints) collection, data pre-processing, and implementation of machine learning algorithms to predict positions based on the collected Wi-Fi fingerprints.

The data collection focuses on access point (AP) information such as SSID, BSSID, and received signal strength

(RSS). The collected fingerprint data undergoes preprocessing to ensure it is clean and consistent for machine learning (ML) training and evaluation. This process is crucial for making accurate predictions. Random Forest (RF) is chosen as the machine learning algorithm [12].

C. PDR Implementation

The proposed scheme uses the PDR technique to estimate the position and orientation of pedestrians by detecting and analyzing the steps taken. The step detection algorithm uses an accelerometer to detect when a step occurs. With the help of a ring buffer, this algorithm calculates the global gravity vector to distinguish between acceleration from steps and other movements. Then, the *z* component of the acceleration will be reduced by the gravity contribution to obtain an estimate of vertical velocity. If this velocity estimate exceeds a certain threshold and occurs after a delay from the previous step, the step is detected.

The orientation of each step is determined using data from the accelerometer and magnetic field sensor. The rotation matrix and orientation angles are calculated to obtain the azimuth value. This value indicates the direction of the pedestrian relative to the geographic north. For the length of each step, the system uses a fixed step length, where the step length in this system is set to a constant 50 cm. This step length is converted to the total travel distance based on the number of detected steps.

V. RESULTS AND ANALYSIS

The experiments were conducted in the New Building of the Faculty of Computer Science, University of Indonesia, from the ground (G) floor up to the 3rd floor. The size of each floor is 67m x 30m. During the offline phase, 182 APs were collected. The RPs are selected according to the Subsection III-B. The preliminary Wi-Fi scanings are conducted on each RP, and the collected APs' RSS are used to train the machine learning model. For system evaluation, the tester walks around the building's G, 1st, 2nd, and 3rd floors carrying the developed Android Application as depicted in Fig. 3. The top-left figure shows the real-time location of user.



Fig. 3. Experiment on 2nd floor of the building

1) *StepThres Value*: To determine the value of *StepThres*, the trend of increasing error produced by PDR is considered. PDR experiments are conducted to determine the location, and the prediction error is collected at each step. Fig. 4 presents the error resulting from PDR as the number of steps increases. It is observed that the error starts to increase significantly from the 25th step. This indicates that after about 25 steps, the accumulation of errors in the PDR method becomes larger and needs correction. Therefore, the *StepThres* is set to 25.

A. Determining Threshold

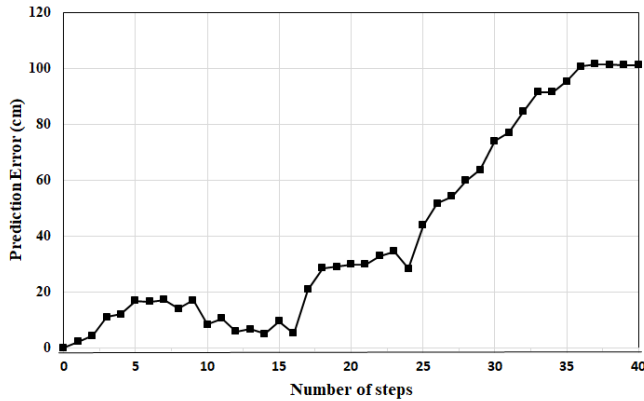


Fig. 4. PDR error trend

1) *DistanceThres Value*: Based on a statistical analysis of PDR error data, an average error of 41.13 cm is found with a standard deviation of about 35.36 cm for 40 steps. Therefore, the *DistanceThres* is set to 2 meters so that this threshold covers most of the error fluctuation that may occur. This threshold helps ensure that position corrections are made only when the Wi-Fi fingerprinting results are sufficiently close to the PDR results.

2) *ScanThres Value*: As the number of steps increases, the resulting error becomes larger, so the correction process using Wi-Fi fingerprinting will not yield a good result. To determine the value of *ScanThres*, the error trend that begins to stagnate at the 36th step is considered. The location correction process using Wi-Fi fingerprinting begins when the user's step equals *StepThres*. From this point, it takes 12 steps to reach the 36th step. Therefore, the attempt threshold for the combined Wi-Fi fingerprinting and PDR algorithm is set at 12 attempts.

B. Fusion Algorithm Results

Fig. 5 illustrates the comparison of coordinate lines generated by the fusion algorithm, standalone PDR, and the actual movement coordinates on the ground floor of the building. The Figure shows that the combined technique produces lines closer to the actual coordinates than those generated by standalone PDR. This indicates that integrating PDR and Wi-Fi Fingerprinting (fusion algorithm) reduces positional errors.

The figure also shows that in the combined technique, some lines cause the movement trace to appear skewed. This occurs due to the use of Wi-Fi prediction results intended to reduce propagation errors from PDR. Despite a slight skewness, the results overall are more accurate than PDR alone. Overall, there is an increase in the accuracy of the fusion algorithm by approximately **55.65%**. It is translated to an average error

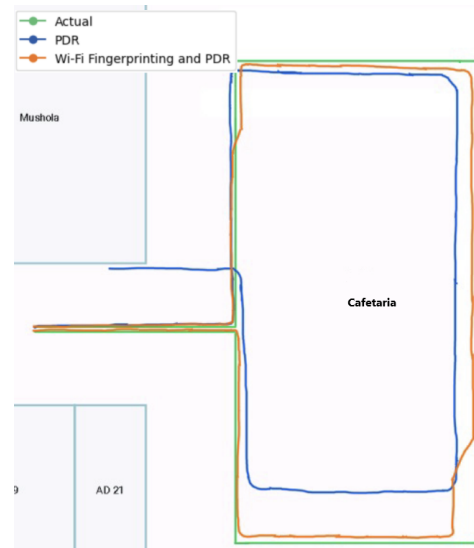


Fig. 5. Comparison of Fusion Algorithm, PDR, and actual path

distance of 0.67 m. Therefore, the use of this combined technique not only improves accuracy but also provides better stability in position tracking.

C. Overhead Analysis

The proposed technique of combining Wi-Fi fingerprinting with PDR can significantly reduce system overhead. Besides its accuracy, the proposed technique has several advantages.

TABLE I
FLOOR PREDICTION MODEL PERFORMANCE

	G Floor	1st Floor	2nd Floor	3rd Floor
Number of RPs	13	18	8	17

1) *Reduced Fingerprint Data Collection*: Table I presents the number of RPs created on each floor for the proposed technique (total 56 grids). The grid size of each RP is 1 m x 1 m. If the conventional Wi-Fi fingerprinting is used where the grid is drawn all over the area, a floor of size 67m x 30m per floor would require $67 \times 30 = 2010$ grids. Thus, there are a total of $2010 \times 4 = 8040$ grids. Assuming that RPs shall be made for 50% of areas when using conventional Wi-Fi fingerprinting, this implies that the proposed technique reduces the number of RPs from 4020 RPs to only 56 RPs. This value translates to a 98.6% overhead reduction. Doubling the total number of RPs of the proposed technique for more accuracy still translates to a 97.2% overhead reduction. Note that a different number of RPs may be required if the indoor structure differs, regardless of the floor size.

2) *Reduce PDR Propagation Error*: The periodic use of Wi-Fi fingerprinting can help adjust and reduce propagation errors in PDR. As shown in Fig. 5, the combined technique helps correct the prediction errors by PDR through periodic adjustments. This makes the user's movement line more accurate than PDR alone.

3) *Save Resources*: Overall, the integration of PDR with Wi-Fi fingerprinting not only reduces effort and increases accuracy but also makes the system more efficient in terms of resource usage such as battery and internet. This is because Wi-Fi scanning is not done continuously. Additionally, the periodic adjustment process performed by Wi-Fi fingerprint-

ing is done at fairly long intervals, which can significantly reduce internet data usage.

VI. CONCLUSION

This work aims to develop and implement an indoor positioning system (IPS) based on collaborative PDR and Wi-Fi fingerprinting. The combination of Wi-Fi fingerprinting and PDR improves accuracy and provides better stability in position tracking, reduces overhead in fingerprint data collection, saves resources like battery and internet data usage, and reduces propagation error from PDR. However, there is still a slight deviation that cannot be avoided due to the characteristics of the Wi-Fi signal itself. The influence of the device used in testing is also quite significant because PDR results can vary depending on the type and brand of the device, as well as the quality and calibration of different inertia sensors. Additionally, variations in step length between users caused by differences in body posture, walking speed, and individual walking styles also affect PDR accuracy. When conducting tests, these factors must be considered to ensure more accurate and consistent results. Those considerations are left for future work.

Finally, this research successfully developed and implemented an IPS utilizing Wi-Fi fingerprinting and PDR. The evaluations are conducted through various scenarios to test the overall system performance. The results of this research contribute to the development of low overhead and highly accurate IPS. Thus, the proposed IPS can be expected to support the indoor LBS in improving indoor user experience through various services.

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